0580_w24_ms_22_Q19

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Partial Marks</th><th>Guidance</th></tr>
<tr><td>19(a)</td><td>858 or 857.7 to 857.9</td><td>3</td><td>

M2 for $\left[\frac{1}{2}\times\right]\frac{4}{3}\times\pi\times4.3^3+\pi\times4.3^2\times11.9$ oe
or **M1** for $\left[\frac{1}{2}\times\right]\frac{4}{3}\times\pi\times4.3^3$
or $\pi\times4.3^2\times11.9$

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Award 3 marks for answers 273.03π to 273.04π but M2 only for 273π (as it is outside the range).
M2 implied by 1024, 326[.0…]π
implied by 333 or 332.8 to 333.2, $\frac{79507}{750}\pi$, 106π
or 167 or 166.4 to 166.6, $\frac{79507}{1500}\pi$, 53π
implied by 691 or 690.8 to 692…, 220π
For **M** marks, in (a) and in (b), allow multiples of π where the multiple is correct to 3sf or better.
For **M1** marks condone ‘picking out’ a term from a string of longer working leading to the answer and ignore names/labelling.

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<tr><td>19(b)</td><td>496 or 495.7 to 495.8…</td><td>4</td><td>

M3 for
$\frac{1}{2}\times4\times\pi\times4.3^2+\pi\times4.3^2+2\times\pi\times4.3\times11.9$
oe
OR
M1 for $\pi\times4.3^2\times k$ where k is a whole number

M1 for $2\times\pi\times4.3\times11.9$

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Award 4 marks for 157.8[1]π or $\frac{15781}{100}\pi$
Award M3 only for answer 158π

Common implied figures for first M1, these or better:
<table>
<tr><th>k =</th><th>3sf</th><th>Range (3.14 to 22/7)</th><th>Exact</th><th>Rounded</th></tr>
<tr><td>1</td><td>58.1</td><td>58.05 to 58.1…</td><td>$\frac{1849}{100}\pi$</td><td>18.5π</td></tr>
<tr><td>2</td><td>116</td><td>116.1 to 116.2…</td><td>$\frac{1849}{50}\pi$</td><td>37π</td></tr>
<tr><td>3</td><td>174</td><td>174.1 to 174.3…</td><td>$\frac{5547}{100}\pi$</td><td>55.5π</td></tr>
<tr><td>4</td><td>232</td><td>232.2 to 232.4…</td><td>$\frac{1849}{25}\pi$</td><td>74π</td></tr>
<tr><td>5</td><td>290</td><td>290.2 to 291</td><td>$\frac{1849}{20}\pi$</td><td>92.5π</td></tr>
<tr><td>6</td><td>349</td><td>348 to 349</td><td>$\frac{5547}{50}\pi$</td><td>111π</td></tr>
<tr><td>7</td><td>407</td><td>406 to 407</td><td>$\frac{12943}{100}\pi$</td><td>129π</td></tr>
</table>
M1 implied by e.g.
321 to 322…, $\frac{5117}{50}\pi$, 102π

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